

COMP0083: Kernel Methods

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Part I

Foundations: Hilbert Spaces and Inner Products

1 Motivation and Overview

Kernels are fundamental tools in machine learning that allow us to:

- Compare objects based on their features without explicitly computing feature representations
- Control model smoothness and generalization to avoid overfitting
- Extend linear methods to non-linear settings through feature space transformations

The key insight is that many algorithms depend only on inner products between data points, which we can compute implicitly through a kernel function.

2 Inner Products and Norms

2.1 Inner Product Spaces

Definition 2.1 (Inner Product). Let H be a vector space over $\mathbb{R}$. A function $\langle \cdot, \cdot \rangle_H : H \times H \rightarrow \mathbb{R}$ is an inner product if it satisfies:

1. **Linearity in the first argument:**

$$\langle \alpha f + \beta g, h \rangle_H = \alpha \langle f, h \rangle_H + \beta \langle g, h \rangle_H$$

for all $\alpha, \beta \in \mathbb{R}$ and $f, g, h \in H$.

2. **Symmetry:** $\langle f, g \rangle_H = \langle g, f \rangle_H$ for all $f, g \in H$.

3. **Positive Definiteness:** $\langle f, f \rangle_H \geq 0$ for all $f \in H$, with equality if and only if $f = 0$.

The inner product induces a norm on H :

$$\|f\|_H := \sqrt{\langle f, f \rangle_H}$$

2.2 Norms on Vector Spaces

Definition 2.2 (Norm). Let F be a vector space over $\mathbb{R}$. A function $\|\cdot\|_F : F \rightarrow [0, \infty)$ is a norm if it satisfies:

1. **Separation of points:** $\|f\|_F = 0 \Leftrightarrow f = 0$
2. **Positive homogeneity:** $\|\lambda f\|_F = |\lambda| \|f\|_F$ for all $\lambda \in \mathbb{R}$ and $f \in F$
3. **Triangle inequality:** $\|f + g\|_F \leq \|f\|_F + \|g\|_F$ for all $f, g \in F$

Every normed vector space carries an induced metric: $d(f, g) = \|f - g\|_F$. Conversely, every inner product space is a normed vector space with the norm induced by the inner product.

2.3 Inner Products on $\mathbb{R}^n$

Definition 2.3 (Inner Product on $\mathbb{R}^n$). A symmetric positive definite matrix $M \in \mathbb{R}^{n \times n}$ defines an inner product on $\mathbb{R}^n$:

$$\langle x, y \rangle_M := x^T M y \quad \text{for all } x, y \in \mathbb{R}^n$$

The special case $M = I$ gives the standard Euclidean inner product: $\langle x, y \rangle = x^T y$.

3 Hilbert Spaces and Completeness

3.1 Cauchy Sequences and Completeness

A sequence $\{f_n\}_{n=1}^\infty$ is Cauchy with respect to the induced metric if for every $\varepsilon > 0$, there exists N such that for all $n, m > N$:

$$\|f_n - f_m\|_F < \varepsilon$$

Every convergent sequence is Cauchy, but not all Cauchy sequences converge. For example, the sequence $1, 1.4, 1.41, 1.414, \dots$ is Cauchy in $\mathbb{Q}$ but converges (to $\sqrt{2}$) only in $\mathbb{R}$.

Definition 3.1 (Complete Metric Space). A normed vector space F is complete if every Cauchy sequence $\{f_n\}$ in F converges to some $f \in F$ (i.e., $\lim_{n \rightarrow \infty} \|f_n - f\|_F = 0$).

Definition 3.2 (Hilbert Space). A Hilbert space is a complete inner product space. That is, it is a vector space with an inner product where every Cauchy sequence converges.

More formally, a Banach space is a complete normed vector space, while a Hilbert space is a complete inner product space.

3.2 Orthogonality and Orthonormal Basis

Definition 3.3 (Orthogonal Complement). For a set $M \subseteq H$ in an inner product space, its orthogonal complement is:

$$M^\perp := \{g \in H : \langle f, g \rangle = 0 \text{ for all } f \in M\}$$

The orthogonal complement is always a closed linear subspace. If M is a linear subspace, then $M \cap M^\perp = \{0\}$ (discrete intersection).

Definition 3.4 (Orthonormal Basis). An orthonormal basis of a Hilbert space H is a family $\{e_k\}_{k \in B}$ of elements satisfying:

1. **Orthogonality:** $\langle e_i, e_j \rangle = 0$ for all $i \neq j$
2. **Normalization:** $\|e_k\|_H = 1$ for all $k \in B$
3. **Completeness:** The linear span of $\{e_k\}$ is dense in H (closedness condition)

Such a basis allows every element $f \in H$ to be represented as:

$$f = \sum_{k \in B} \langle f, e_k \rangle e_k$$

with convergence in the Hilbert space norm.

3.3 Important Properties in Inner Product Spaces

Theorem 3.1 (Cauchy-Schwarz Inequality). For any elements f, g in an inner product space, we have:

$$|\langle f, g \rangle| \leq \|f\| \|g\|$$

with equality if and only if f and g are linearly dependent.

Theorem 3.2 (Pythagorean Theorem). If $f \perp g$ (i.e., $\langle f, g \rangle = 0$), then:

$$\|f + g\|^2 = \|f\|^2 + \|g\|^2$$

Theorem 3.3 (Parallelogram Law).

$$2\|f\|^2 + 2\|g\|^2 = \|f + g\|^2 + \|f - g\|^2$$

4 Linear Operators

4.1 Bounded and Continuous Operators

Definition 4.1 (Linear Operator). A map $A : F \rightarrow G$ between vector spaces is a linear operator if:

$$A(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 A(f_1) + \alpha_2 A(f_2)$$

for all scalars α_1, α_2 and elements $f_1, f_2 \in F$.

When $G = \mathbb{R}$ (or $\mathbb{C}$), a linear operator is called a functional.

Definition 4.2 (Operator Norm). The operator norm of a bounded linear operator $A : F \rightarrow G$ is:

$$\|A\| := \sup_{f \in F, f \neq 0} \frac{\|Af\|_G}{\|f\|_F} = \sup_{\|f\|_F \leq 1} \|Af\|_G$$

If $\|A\| < \infty$, we say A is a bounded linear operator.

Theorem 4.1 (Equivalence of Boundedness and Continuity). For a linear operator $A : F \rightarrow G$ between normed spaces, the following are equivalent:

1. A is a bounded operator
2. A is continuous on all of F
3. A is continuous at a single point in F

4.2 The Riesz Representation Theorem

Theorem 4.2 (Riesz Representation Theorem). Let H be a Hilbert space and let $f : H \rightarrow \mathbb{R}$ be a bounded linear functional. Then there exists a unique $y \in H$ such that:

$$f(x) = \langle x, y \rangle_H \quad \text{for all } x \in H$$

Moreover, $\|f\|_{H'} = \|y\|_H$, where $\|f\|_{H'}$ is the operator norm of f on the dual space.

Remark 4.1. This theorem is fundamental: it says that every bounded linear functional on a Hilbert space can be represented as an inner product with some fixed vector. The mapping $J : H \rightarrow H'$ defined by $J(y)(x) = \langle x, y \rangle_H$ is an isometric isomorphism between H and its dual space H' .

Part II

Kernels and Reproducing Kernel Hilbert Spaces

5 Kernels: Definition and Properties

5.1 Kernel Definition and Positive Definiteness

Definition 5.1 (Kernel). Let X be a non-empty set. A function $k : X \times X \rightarrow \mathbb{R}$ is a kernel if there exists a Hilbert space H and a feature map $\phi : X \rightarrow H$ such that:

$$k(x, x') = \langle \phi(x), \phi(x') \rangle_H \quad \text{for all } x, x' \in X$$

Equivalently, a kernel can be characterized through positive definiteness.

Definition 5.2 (Positive Definite Kernel). A function $k : X \times X \rightarrow \mathbb{R}$ is positive definite if for any finite set of points $\{x_1, \dots, x_n\} \subseteq X$ and coefficients $\{c_1, \dots, c_n\} \subseteq \mathbb{R}$:

$$\sum_{i,j=1}^n c_i c_j k(x_i, x_j) \geq 0$$

The key insight is that kernels and positive definite functions are equivalent concepts.

5.2 Constructing Kernels

Proposition 5.1 (Operations Preserving Kernels). The following operations preserve the kernel property:

1. **Positive scaling:** If k is a kernel and $\alpha > 0$, then αk is a kernel.
2. **Sums of kernels:** If k_1 and k_2 are kernels on X , then $k_1 + k_2$ is a kernel on X .
3. **Products of kernels:** If k_1 is a kernel on X_1 and k_2 is a kernel on X_2 , then the product

$$k(x, x') := k_1(x_1, x'_1) \cdot k_2(x_2, x'_2)$$

is a kernel on $X_1 \times X_2$.

4. **Kernel composition:** If $A : X \rightarrow \tilde{X}$ is a map and $\tilde{k}$ is a kernel on $\tilde{X}$, then $k(x, x') := \tilde{k}(A(x), A(x'))$ is a kernel on X .

Remark 5.1. Not all differences of kernels are kernels. Additionally, a single kernel can admit infinitely many different feature map representations.

5.3 Examples of Kernels

Definition 5.3 (Polynomial Kernel). The polynomial kernel of degree d is defined as:

$$k(x, x') := (\langle x, x' \rangle + c)^d$$

where d is a positive integer and $c \geq 0$ is a constant. This kernel corresponds to the feature space of all monomials of degree up to d in the original features.

Definition 5.4 (Gaussian (RBF) Kernel). The Gaussian or Radial Basis Function (RBF) kernel is:

$$k(x, x') := \exp(-\gamma \|x - x'\|^2)$$

where $\gamma > 0$ is the bandwidth parameter. This kernel is universal and characteristic for probability measures.

Definition 5.5 (Kernels from Power Series). For $r \in (0, \infty]$ and coefficients $\{a_n\}_{n=0}^{\infty}$ with $a_n \geq 0$, if the radius of convergence allows, we can define:

$$k(x, x') := \sum_{n=0}^{\infty} a_n \langle x, x' \rangle^n$$

for $\langle x, x' \rangle \in (-r, r)$.

6 Reproducing Kernel Hilbert Spaces (RKHS)

6.1 Evaluation Functionals

Definition 6.1 (Evaluation Functional). Given a space H of functions $f : X \rightarrow \mathbb{R}$ and a fixed point $x_0 \in X$, the evaluation functional at x_0 is:

$$E_{x_0} : H \rightarrow \mathbb{R}, \quad E_{x_0}(f) := f(x_0)$$

This is a linear functional, but it is not always bounded with respect to every norm on H .

6.2 RKHS Definition

Definition 6.2 (Reproducing Kernel Hilbert Space). A Hilbert space H of functions $f : X \rightarrow \mathbb{R}$ is a Reproducing Kernel Hilbert Space (RKHS) if for every $x \in X$, the evaluation functional $E_x : H \rightarrow \mathbb{R}$ defined by $E_x(f) = f(x)$ is bounded.

Equivalently, there exists a family $\{k_x\}_{x \in X} \subseteq H$ such that:

$$f(x) = \langle f, k_x \rangle_H \quad \text{for all } x \in X, f \in H$$

where $k_x(y) = k(y, x)$ and $k : X \times X \rightarrow \mathbb{R}$ is the reproducing kernel.

This definition encodes the **reproducing property**: function values can be recovered via inner products with kernel sections.

6.3 Moore-Aronszajn Theorem

Theorem 6.1 (Moore-Aronszajn). There is a one-to-one correspondence between positive definite kernels on $X \times X$ and Reproducing Kernel Hilbert Spaces on X .

Specifically, for every positive definite kernel k on $X \times X$, there exists a unique RKHS H_k with reproducing kernel k . Conversely, every RKHS has a unique reproducing kernel.

The RKHS H_k is the completion of the pre-Hilbert space spanned by $\{k_x\}_{x \in X}$ with inner product:

$$\left\langle \sum_i \alpha_i k_{x_i}, \sum_j \beta_j k_{x_j} \right\rangle_{H_k} := \sum_{i,j} \alpha_i \beta_j k(x_i, x_j)$$

Key consequences:

- Every positive definite kernel defines a unique RKHS
- The feature map $\phi(x) = k(\cdot, x)$ embeds X into H_k
- The kernel value is $k(x, x') = \langle \phi(x), \phi(x') \rangle_{H_k}$

6.4 Properties of RKHS and Smoothness

Proposition 6.1 (Smoothness from Small RKHS Norm). If $f \in H_k$ with small $\|f\|_{H_k}$, then f is smooth. Specifically:

$$|f(x) - f(x')| = |\langle f, k_x - k_{x'} \rangle| \leq \|f\|_{H_k} \|k_x - k_{x'}\|_{H_k}$$

For smooth kernels, $\|k_x - k_{x'}\|_{H_k}$ is small when x is close to x' .

7 Spectral Properties and Function Representations

7.1 Integral Linear Operators and Mercer's Theorem

Definition 7.1 (Integral Operator). Given a kernel k and a measure ν on X , the integral operator is:

$$(T_k f)(x) := \int_X k(x, y) f(y) d\nu(y)$$

This is a linear operator mapping appropriate function spaces.

Theorem 7.1 (Mercer's Theorem). Let k be a continuous kernel on a compact metric space X , and let ν be a finite Borel measure with $\text{supp}(\nu) = X$. Then k admits an eigenfunction expansion:

$$k(x, y) = \sum_{j=1}^{\infty} \lambda_j e_j(x) e_j(y)$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ are eigenvalues of the integral operator T_k and $\{e_j\}$ are the corresponding orthonormal eigenfunctions. The series converges absolutely and uniformly on $X \times X$.

7.2 Spectral Theorem and Decomposition

Theorem 7.2 (Spectral Decomposition). Let H be a Hilbert space and $T : H \rightarrow H$ a compact, self-adjoint operator. Then there exists an at-most countable orthonormal set $\{e_j\}_{j \in J}$ and eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ converging to zero, such that:

$$f = \sum_{j \in J} \langle f, e_j \rangle e_j \quad \text{and} \quad Tf = \sum_{j \in J} \lambda_j \langle f, e_j \rangle e_j$$

for all $f \in H$.

This allows us to represent any function in H_k and understand the decay of the spectrum.

Part III

Applications of Kernels

8 Kernel Methods in Machine Learning

8.1 Representer Theorem

Theorem 8.1 (Representer Theorem). Consider a regularized empirical risk minimizer:

$$\min_{f \in H_k} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \|f\|_{H_k}^2$$

where L is any loss function and $\lambda > 0$. The optimal solution has the form:

$$f^*(x) = \sum_{i=1}^n \alpha_i^* k(x, x_i)$$

That is, the optimal function is expressible as a linear combination of kernel evaluations at the training points.

This is why kernel methods are computationally practical: we only need to optimize over n coefficients rather than solving in infinite-dimensional spaces.

8.2 Kernel Ridge Regression

Definition 8.1 (Kernel Ridge Regression). Kernel Ridge Regression solves:

$$\min_{f \in H_k} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{H_k}^2$$

By the representer theorem, the solution is:

$$f^*(x) = \sum_{i=1}^n \alpha_i^* k(x, x_i)$$

where $\alpha^* = (K + n\lambda I)^{-1} \mathbf{y}$ and K is the kernel matrix with entries $K_{ij} = k(x_i, x_j)$.

9 Kernel Mean Embeddings

9.1 Definition and Properties

Definition 9.1 (Kernel Mean Embedding). For a probability measure P on $\mathcal{X}$ and kernel k , the kernel mean embedding is:

$$\mu_P := \mathbb{E}_{X \sim P}[k(\cdot, X)] = \int k(\cdot, x) dP(x) \in H_k$$

For an empirical measure $P_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, the empirical mean embedding is:

$$\hat{\mu}_{P_n} := \frac{1}{n} \sum_{i=1}^n k(\cdot, x_i)$$

Mean embeddings map probability distributions to function space, preserving distributional information.

Proposition 9.1 (Injectivity for Characteristic Kernels). A kernel k is called characteristic for probability measures on $\mathcal{X}$ if the embedding $P \mapsto \mu_P$ is injective. That is:

$$\mu_P = \mu_Q \implies P = Q$$

The Gaussian kernel and many translation-invariant kernels with full support are characteristic.

10 Maximum Mean Discrepancy (MMD)

10.1 Definition

Definition 10.1 (Maximum Mean Discrepancy). The MMD between two probability measures P and Q with a kernel k is:

$$\text{MMD}_k(P, Q) := \|\mu_P - \mu_Q\|_{H_k}$$

Equivalently:

$$\text{MMD}_k^2(P, Q) = \mathbb{E}_{x, x' \sim P}[k(x, x')] - 2\mathbb{E}_{x \sim P, y \sim Q}[k(x, y)] + \mathbb{E}_{y, y' \sim Q}[k(y, y')]$$

MMD provides a metric on the space of probability measures via an RKHS.

10.2 Witness Function and Interpretation

The function that witnesses the maximum discrepancy—the one that achieves the supremum in the definition—is the normalized difference of mean embeddings:

$$f^*(x) = \frac{\mu_P(x) - \mu_Q(x)}{\|\mu_P - \mu_Q\|_{H_k}}$$

This function shows where the distributions differ most in the feature space.

10.3 Two-Sample Testing with MMD

Proposition 10.1 (Hypothesis Testing). Under the null hypothesis $H_0 : P = Q$, the statistic $\text{MMD}_k^2(P_n, Q_m)$ converges to 0. The asymptotic distribution can be derived for test construction.

For large samples and characteristic kernels, MMD provides a consistent test for distributional differences.

11 Hilbert-Schmidt Independence Criterion (HSIC)

11.1 Constrained Covariance

Definition 11.1 (Constrained Covariance). For two probability measures P on spaces with kernels k and ℓ , the constrained covariance is:

$$\text{COCO}(P) := \sup_{\|f\|_{H_k}=1, \|g\|_{H_\ell}=1} \text{Cov}[f(X), g(Y)]$$

where $\text{Cov}[f(X), g(Y)] = \mathbb{E}[f(X)g(Y)] - \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$.

11.2 HSIC Definition

Definition 11.2 (Hilbert-Schmidt Independence Criterion). The HSIC between random variables X (with kernel k) and Y (with kernel ℓ) is:

$$\text{HSIC}(P_{XY}; H_k, H_\ell) := \|\mathcal{C}_{XY}\|_{HS}^2$$

where $\mathcal{C}_{XY}$ is the cross-covariance operator in the tensor product Hilbert space.

HSIC measures statistical dependence and is zero if and only if $X \perp Y$.

11.3 Empirical HSIC

For samples $\{(x_i, y_i)\}_{i=1}^n$, the empirical HSIC can be computed efficiently as:

$$\widehat{\text{HSIC}} = \frac{1}{n^2} \text{tr}(KHLH)$$

where K and ℓ are the kernel matrices and $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the centering matrix.

Part IV

Advanced Topics

12 Kernel Stein Discrepancy

12.1 Stein Operators

Definition 12.1 (Langevin Stein Operator). For a probability distribution p with score function $\nabla \log p(x)$, the Langevin Stein operator is:

$$(\mathcal{A}_p f)(x) := \nabla \log p(x) \cdot f(x) + \nabla f(x)$$

This operator has the key property that:

$$\mathbb{E}_p[\mathcal{A}_p f] = 0$$

for all appropriate test functions f .

12.2 Definition of KSD

Definition 12.2 (Kernel Stein Discrepancy). The Kernel Stein Discrepancy (KSD) measures the maximum violation of the Stein equation:

$$\text{KSD}(Q, p; H_k) := \sup_{\|f\|_{H_k} \leq 1} |\mathbb{E}_{X \sim Q}[(\mathcal{A}_p f)(X)]|$$

KSD is zero if and only if $Q = p$, making it a proper divergence for distribution testing without density ratio estimation.

KSD is particularly useful when we have an unnormalized density or samples from a distribution but cannot compute the likelihood.

13 Optimization and Convex Analysis

13.1 Lagrangian and Duality

Definition 13.1 (Lagrangian). For a constrained optimization problem:

$$\min_x f_0(x) \quad \text{subject to } f_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, p$$

the Lagrangian is:

$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

where $\lambda_i \geq 0$ are inequality constraint multipliers and ν_j are equality constraint multipliers.

The dual problem maximizes:

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu)$$

subject to $\lambda \geq 0$.

13.2 Strong Duality and KKT Conditions

Theorem 13.1 (Strong Duality via Slater's Condition). If the primal problem is convex and Slater's condition holds (there exists a feasible point with strict inequality for inequality constraints), then strong duality holds: the optimal value of the primal equals the optimal value of the dual.

Theorem 13.2 (KKT Conditions). A point x^* with multipliers (λ^*, ν^*) satisfies the KKT conditions if:

1. **Stationarity:** $0 \in \partial f_0(x^*) + \sum_i \lambda_i^* \partial f_i(x^*) + \sum_j \nu_j^* \partial h_j(x^*)$
2. **Complementary slackness:** $\lambda_i^* f_i(x^*) = 0$ for all i
3. **Primal feasibility:** $f_i(x^*) \leq 0$ and $h_j(x^*) = 0$
4. **Dual feasibility:** $\lambda_i^* \geq 0$ for all i

For convex problems with strong duality, the KKT conditions characterize optimal solutions.

Complementary slackness is key: if $\lambda_i^* > 0$, then $f_i(x^*) = 0$ (the constraint is active).